UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 11

Content.

1. Properties of Sub-Gaussian Distributions
2. Sub-Exponential Distributions

1 Sub-Gaussian Distributions Continued

Theorem 1. *(Sub-Gaussian Properties)*
Let X be a random variable. Then, the following are equivalent, where the constants $K_i > 0$ differ from each other by constant factors.

1. (Tails) For any $t \geq 0$, $\mathbb{P}[|X| > t] \leq 2\exp(-\frac{t^2}{K_1})$.
2. (Moments) For any $p \geq 1$, $\|x\|_{L_p} := (\mathbb{E}|X|^p)^{\frac{1}{p}} \leq K_2\sqrt{p}$.
3. (MGF of X^2) For all $|\lambda| \leq \frac{1}{K_3}$, $\mathbb{E}[e^{\lambda^2 x^2}] \leq \exp(K_3\lambda^2)$.
4. (Boundedness) For some K_4, $\mathbb{E}[\exp(\frac{x^2}{K_4^2})] \leq 2$.

If, additionally, $\mathbb{E}X = 0$, then the above are also equivalent to

5. (MGF of X) For all $\lambda \in \mathbb{R}$, $\mathbb{E}[e^{\lambda X}] \leq \exp(K_5\lambda^2)$.

Remark. If X is sub-Gaussian, then the MGF of X^2 is dominated by a sub-Gaussian MGF, though on an interval enclosing 0.

To prove the above result, we need a lemma from STAT 553, whose proof is a direct application of Fubini-Tonelli theorem.

Lemma 1. $\mathbb{E}|X| = \int_0^\infty \mathbb{P}[|X| \geq t]\, dt$.

Proof of Lemma 1. Note that

$$\int_0^\infty \mathbb{P}[|X| \geq t]\, dt = \int_0^\infty \int_\Omega \mathbb{1}_{|X(\omega)|\geq t}\, d\mathbb{P}(\omega)\, dt = \int_\Omega \int_0^\infty \mathbb{1}_{|X(\omega)|\geq t}\, dt\, d\mathbb{P}(\omega)$$

by Fubini-Tonelli theorem. This completes the proof since

$$\int_0^\infty \mathbb{1}_{|X(\omega)|\geq t}\, dt = |X(\omega)|.$$

□

Proof of the Theorem 1. We first show 1 implies 2. Assume WLOG that $K_1 = 1$.We have

$$\begin{aligned}
\mathbb{E}|X|^p &= \int_0^\infty \mathbb{P}[|X|^p \geq u]\, du \\
&= \int_0^\infty \mathbb{P}[|X| \geq t] p t^{p-1}\, dt && \text{(change of variable } u = t^p\text{)} \\
&\leq 2\int_0^\infty e^{-t^2} p t^{p-1}\, dt && \text{(by condition in 1)} \\
&= \int_0^\infty e^{-s} p s^{\frac{p-2}{2}}\, ds && \text{(change of variable } t^2 = s\text{)} \\
&= p\Gamma(\frac{p}{2}),
\end{aligned}$$

where $\Gamma(s)$ is the Gamma function. Now

$$\begin{aligned}
\mathbb{E}|X|^p &\leq p\Gamma(\frac{p}{2}) \\
&\leq 3p(\frac{p}{2})^{\frac{p}{2}}. && (\Gamma(x) \leq 3x^x \text{ if } x \geq \frac{1}{2})
\end{aligned}$$

So

$$\begin{aligned}
\|X\|_{L_p} &\leq (3p)^{\frac{1}{p}}\sqrt{\frac{p}{2}} \\
&\leq \frac{3e^{\frac{1}{e}}}{\sqrt{2}}\sqrt{p} && (3^{\frac{1}{p}} \leq 3, p^{\frac{1}{p}} \leq e^{\frac{1}{e}}) \\
&=: K_2\sqrt{p}.
\end{aligned}$$

We now show 2 implies 3. As before, assume $K_2 = 1$. We have

$$\begin{aligned}
\mathbb{E}[e^{\lambda^2 x^2}] &= \mathbb{E}[1 + \sum_{k=1}^{\infty} \frac{(\lambda^2 x^2)^k}{k!}] \\
&= 1 + \sum_{k=1}^{\infty} \mathbb{E}[\frac{(\lambda^2 x^2)^k}{k!}] && \text{(monotone convergence theorem)} \\
&= 1 + \sum_{k=1}^{\infty} \frac{\lambda^{2k} \mathbb{E}[x^{2k}]}{k!} \\
&\leq 1 + \sum_{k=1}^{\infty} \frac{\lambda^{2k} (2k)^k}{k!} && \text{(by condition in 2)} \\
&\leq 1 + \sum_{k=1}^{\infty} \frac{\lambda^{2k} (2k)^k}{(\frac{k}{e})^k} && (k! \geq (\frac{k}{e})^k) \\
&= \sum_{k=0}^{\infty} (2\lambda^2 e)^k \\
&= \frac{1}{1 - 2\lambda^2 e} && \text{(valid if } 2\lambda^2 e < 1) \\
&\leq e^{4e\lambda^2} && (\frac{1}{1-x} \leq e^{2x} \text{ if } x \in [0, \frac{1}{2}]) \\
&=: e^{K_3 \lambda^2}.
\end{aligned}$$

Note the final inequality is valid if $2\lambda^2 e \leq \frac{1}{2}$, which is $|\lambda| \leq \frac{1}{2\sqrt{e}}$.

For 3 implying 4, notice that in 3, the upper bound tends to 1 as λ tends to 0, so the condition in 4 can be satisfied as long as K_4 is large enough.

Next we show 4 implies 1. Take $K_4 = 1$ as usual. Just like what we did in showing Chernoff's inequality, we have

$$\begin{aligned}
\mathbb{P}[|X| \geq t] &= \mathbb{P}[e^{X^2} \geq e^{t^2}] \\
&\leq e^{-t^2} \mathbb{E}[e^{X^2}] \\
&\leq 2e^{-t^2} && \text{(by condition in 4)} \\
&=: 2e^{-\frac{t^2}{K_1}}.
\end{aligned}$$

It remains to consider 5. Assume $\mathbb{E}X = 0$. We want to show that 3 implies 5, and 5 implies 1. For the first direction, WLOG. let $K_3 = 1$, we get that if

$|\lambda| \leq 1$,

$$\begin{aligned}
\mathbb{E}[e^{\lambda X}] &\leq \mathbb{E}[\lambda X + E^{\lambda^2 X^2}] && (e^x \leq x + e^{x^2}) \\
&= \mathbb{E}[e^{\lambda^2 X^2}] \\
&\leq e^{\lambda^2}
\end{aligned}$$

If $|\lambda| > 1$, we have

$$\begin{aligned}
\mathbb{E}[e^{\lambda X}] &\leq \mathbb{E}[e^{\frac{\lambda^2 + X^2}{2}}] && (a^2 + b^2 \geq 2ab) \\
&= e^{\frac{\lambda^2}{2}} \mathbb{E}[e^{\frac{X^2}{2}}] \\
&\leq e^{\frac{\lambda^2 + 1}{2}} && \text{(by condition in 3 on } \mathbb{E}[e^{\frac{X^2}{2}}]) \\
&\leq e^{\lambda^2}. && (|\lambda| > 1)
\end{aligned}$$

For the second direction, by the same Chernoff trick,

$$\begin{aligned}
\mathbb{P}[X \geq t] &= \mathbb{P}[e^{\lambda X} \geq e^{\lambda t}] \\
&\leq e^{-\lambda t} \mathbb{E}[e^{\lambda X}] \\
&\leq e^{-\lambda t} e^{\lambda^2}. && \text{(by condition in 5, assuming WLOG. that } K_5 = 1)
\end{aligned}$$

Optimizting over $\lambda > 0$, we get a desired one-sided bound. Steps for getting the result for the other side are similar. □

Definition 1. Given a sub-Gaussian random variable X, we define its sub-Gaussian *norm* by

$$\|X\|_{\psi_2} := \inf\{t > 0 : \mathbb{E} \exp(\frac{X^2}{t^2}) \leq 2\}.$$

Exercise 1. Show that $\| \cdot \|_{\psi_2}$ is indeed a norm in the space of sub-Gaussian random variable.

Remark. (Operational Meaning of $\| \cdot \|_{\psi_2}$) $\|X\|_{\psi_2}$ is the smallest number such that up to constants, we have $\mathbb{E} \exp(\frac{X^2}{\|X\|_{\psi_2}^2}) \leq 2$, $\mathbb{P}[|X| \geq t] \leq 2 \exp(-\frac{-c_1 t^2}{\|X\|_{\psi_2}})$, and so on.

2 Sub-Exponential Distributions

To motivate the definition of sub-exponential distributions and get some flavor of their behavior, we consider two examples.

Example : Note that sub-Gaussian property fails to capture some important distributions. For example, the square of a sub-Gaussian variable X. This is because typically $\mathbb{P}[X^2 > t] = \mathbb{P}[|X| > \sqrt{t}] \leq \exp(-ct)$.

Example : Another example with such tails is $X \sim \text{Lap}(1)$, where $\mathbb{P}[|X| > t] = e^{-\frac{1}{t}}$. In this case, we have $\mathbb{E}[e^{sX}] = \frac{1}{1-s^2}$, $|s| \leq 1$. Moreover, when $|s| \leq \frac{1}{\sqrt{2}}$, we get $\mathbb{E}[e^{sX}] \leq e^{2s^2}$ (since $\frac{1}{1-x} \leq e^{2x}$ when $x \in [0, \frac{1}{2}]$), which means that in some interval containing 0, the MGF is dominated by that of a sub-Gaussain variable. But outside, the MGF blows up.

Definition 2. A random variable with mean μ is sub-exponential if there exists a pair of positive numbers (σ^2, α) such that

$$\mathbb{E}[e^{s(X-\mu)}] \leq e^{\frac{s^2\sigma^2}{2}}$$

for all $|s| < \frac{1}{\alpha}$.

Theorem 2. Suppose that X with mean μ is sub-exponential with (σ^2, α). Then

$$\mathbb{P}[X - \mu \geq t] \leq \begin{cases} e^{-\frac{t^2}{2\sigma^2}}, & t \in [0, \frac{\sigma^2}{\alpha}] \\ e^{-\frac{t}{2\alpha}}, & t \in [\frac{\sigma^2}{\alpha}, \infty) \end{cases}$$

Remark. Here are two kinds of tails. For large deviations, it's controlled by exponential tail.

Proof. Still, by the similar technique as in Chernoff bound, we get

$$\mathbb{P}[X - \mu \geq t] \leq e^{-\lambda t}\mathbb{E}[e^{\lambda(X-\mu)}] \leq e^{-\lambda t + \frac{\lambda^2\sigma^2}{2}},$$

where $0 < \lambda \leq \frac{1}{\alpha}$. Now we need to optimize over λ within this constraint, and we will get the exact bound after doing this. □

Corollary 1.

$$\mathbb{P}[|X - \mu| \geq t] \leq 2e^{-\frac{t^2}{2\sigma^2 + 2\alpha t}}.$$

The proof of this corollary is by combining the two parts in the above theorem together by using $\min\{\frac{1}{a}, \frac{1}{b}\} \geq \frac{1}{a+b}$ for $a, b > 0$.

Lemma 2. Let $X_i \sim \text{subExp}(\sigma^2, \alpha_i)$, $i \in \{1, ..., n\}$ be independent sub-exponential random variables all with mean 0, then

$$\sum_{i=1}^{n} X_i \sim \text{subExp}(\sum_i \sigma_i^2, \max_i \alpha_i),$$

which means sub-exponential families are closed under convolution.

Proof.

$$\begin{aligned}
\mathbb{E}[e^{s\sum_i X_i}] &= \prod_i \mathbb{E}[e^{sX_i}] \\
&\leq \prod_i e^{\frac{s^2\sigma_i^2}{2}} && \text{when } |s| < \min_i \frac{1}{\alpha_i} \\
&= e^{\frac{s^2\sum_i \sigma_i^2}{2}},
\end{aligned}$$

. □